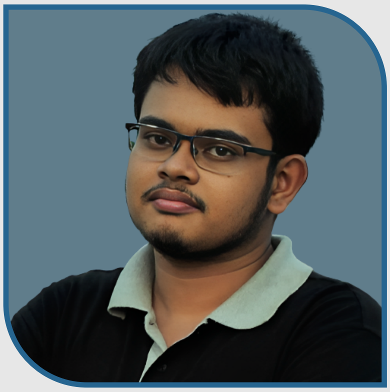

Md Mahbub Ali

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Bangladeshi

Social Network

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Language

English - Fluent

Skills

- PSCAD
- MATLAB Simulink
- PowerFactory
- Python - basic level
- MS Office, Google Editors

About Me

Currently working on a thesis focused on VSG-based grid-forming converter control systems.
I am actively seeking job opportunities where I can apply my knowledge, skills, gain hands-on experience, and contribute to the power system industry. My goal is to become a grid engineer, and I am deeply passionate about renewable energy and power systems. However, I am open to exploring any role related to electrical power systems.

Thesis Work (Masters Study)

- **Synchronization and Fault Analysis of VSG-Based Grid-Forming Inverter with a Synchronous Generator**
  - Simulating grid synchronization between a virtual synchronousgenerator (VSG)-based inverter and a conventional synchronous generator in PSCAD.
  - Analyzing variations in inertia and damping to assess dynamic performance of low-inertia systems under disturbance.

Academic Projects (Masters Study)

- **Black-Start Capability Analysis via Grid Forming Control** (3rd Semester Project)
  - Studied droop-based grid-forming control system application.
  - Simulated an 80MVA single-line diagram in PSCAD using open source data from Eurowind Energy Company to supply a 20MW/15MVAr load 30s post-blackout via a grid-forming converter.
- **Machine Learning Based Power Flow Analysis** (2nd Semester Project)
  - Studied the use of Graph Neural Networks to optimize Newton-Raphson-based power flow analysis, achieving faster computation compared to traditional methods. Utilized the basic understanding of Python.
  - Validated machine learning analysis data using PowerFactory.
- **Fractional Frequency Transmission System for Offshore Wind Power** (1st Semester Project)
  - Designed and simulated of a three-phase cyclo converter for bidirectional frequency conversion using Simulink.

Internship

- **Embedded Electronics Internship (Jan,2022 - Jun,2022):** Aqualink Bangladesh Limited
  - Developed protective circuits, digital and analog control circuits for advanced electronic systems.

Education

Current(4th Semester) — **MSc. in Energy Engineering** — Aalborg University

- Studied power system stability, advanced power electronics, high voltage system, electrical machines, and control systems, windpower system, and battery energy storage system.

Jan,2016 - Dec,2020 — **BSc. in Electrical and Electronic Engineering** — BRAC University

- Studied power plant engineering, energy conversion, and power electronics.
- **Bachelor Thesis:** Conducted a theoretical energy accumulation study on a two-level dual-axis bi-facial module-based PV system using MATLAB-code.